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# Yonatan Konstantine

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## EDUCATION

**Worcester Polytechnic Institute (WPI)** **Aug. 2022 — May 2027**
B.S. & M.S. in Electrical and Computer Engineering *Worcester, MA*

**Relevant Courses**
Machine Learning, Signal Analysis, Information Theory, Embedded Systems, Digital System Design, Microelectronics

## SKILLS

**Programming/HDL:** C, Verilog, MATLAB, Java
**Hardware:** STM32, Arduino, MSP430, FPGA, GPIO, PWM, Embedded Systems, Analog & Digital Circuit Design
**Electronics & Testing:** Oscilloscopes, Digital Multimeters, Breadboard Prototyping, Circuit Debugging, Soldering
**Software & Tools**: LTspice, Vivado, NI Multisim, VS Code
**Languages:** English, Hebrew

## WORK EXPERIENCE

**IT Service Desk Specialist** **May 2023 — Present**
*Worcester Polytechnic Institute* *Worcester, MA*

- Triage, troubleshoot, and resolve 10,000+ technical support cases involving hardware, software, networking, account, and system issues for students, faculty, and staff.
- Collaborate with Network and Systems Operations teams to resolve complex incidents and develop technical documentation that improves support efficiency and user self-service.

## PROJECT EXPERIENCE

**Major Qualifying Project — Project Early Bird** **Aug. 2025 — Apr. 2026**
*Worcester Polytechnic Institute* *Worcester, MA*

- Designed and prototyped a directional parametric speaker using ~40 kHz ultrasonic transducers to produce highly focused audible sound through ultrasonic modulation.
- Developed and iteratively tested embedded and analog driver architectures, integrating STM32 and Arduino microcontrollers, MOSFET switching, and ultrasonic transducer arrays.
- Redesigned and debugged multiple hardware iterations, resolving component, wiring, voltage-rating, signal-generation, and legacy hardware/software compatibility issues to progress from nonfunctional prototypes to controlled audible output.
- Implemented and experimentally tuned a resonant piezoelectric drive circuit, using the transducer's capacitance and discrete inductance to improve drive voltage and efficiency near its ~40 kHz resonant frequency.
- Validated system performance through circuit measurements, acoustic testing, and 360° directivity measurements, alongside SolidWorks modeling and ANSYS modal, harmonic, and acoustic simulations.

**Project Smartzle** **Oct. 2024 – Dec. 2024**
*Worcester Polytechnic Institute* *Worcester, MA*

- Developed an interactive electronic puzzle prototype integrating an Arduino Nano, GPIO expander, LCD display, and conductive game pieces.
- Designed and prototyped hardware/software interactions to create responsive puzzle mechanics while incorporating sustainable design principles.

## LEADERSHIP EXPERIENCE

**A Cappella, Music Director** **Feb. 2024 — Feb. 2026**
*Audiophiles, Worcester Polytechnic Institute* *Worcester, MA*

- Led rehearsals, competed in the International Competition of Collegiate A Cappella (ICCA)
- Mixed, mastered, and released extended play *BPM* onto streaming platforms